

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-4040 - Digital Electronics
Spring Term 2026

Lab. 10-ECL Noise Margins and Fan Out

Issued: 4/8/2026

Due: 4/10/2026

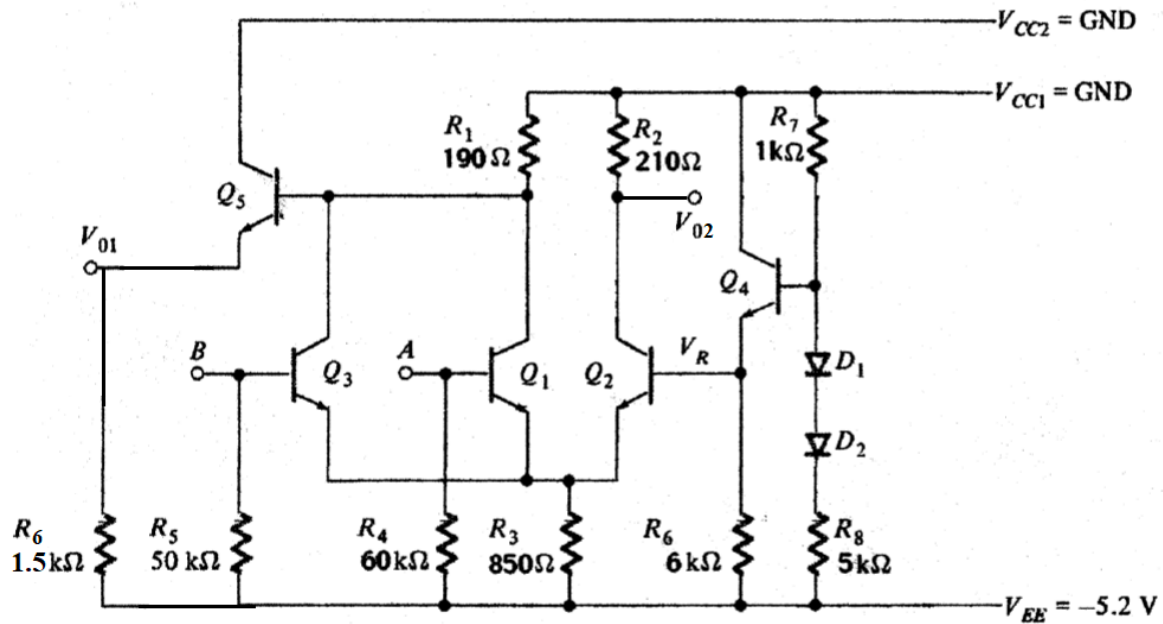
Question 1: Based on Problems 7.9 - 7.12 (D.A. Hodges and H.G. Jackson, *Analysis and Design of Digital Integrated Circuits*, 2nd Ed., McGraw-Hill, 1988.)

Objectives:

- BJT ECL Noise Margin calculations

Given the NPN BJT below where:

- $\beta_F = 80$
- $V_{BE}(FA) = 0.7V$
- $V_{BE}(sat) = 0.7V$
- $V_{CE}(sat) = 0.00V$
- $V_D(on) = 0.7V$



- a) Find V_{OH} , V_{OL} , V_{IH} , V_{IL} for when one input switches, and the other is logic low. Solving for either provides the same result. Find V_{OH} and V_{OL} for both V_{O1} and V_{O2} . Both outputs will have the same V_{IL} and V_{IH} points.

- i. First find V_{OH} and V_{OL}
 - 1) For logic low input, assume that logic 0 = $V_R - 0.35V$.
 - 2) For logic high input, assume that logic 1 = $V_R + 0.35V$.
- ii. Once V_{OL} is known, V_{SWING} can be solved for. V_{IH} and V_{IL} points can be found with the following equations:

$$V_{IH} \cong V_R + V_T \ln\left(\frac{V_{SWING}}{V_T} - 2\right)$$

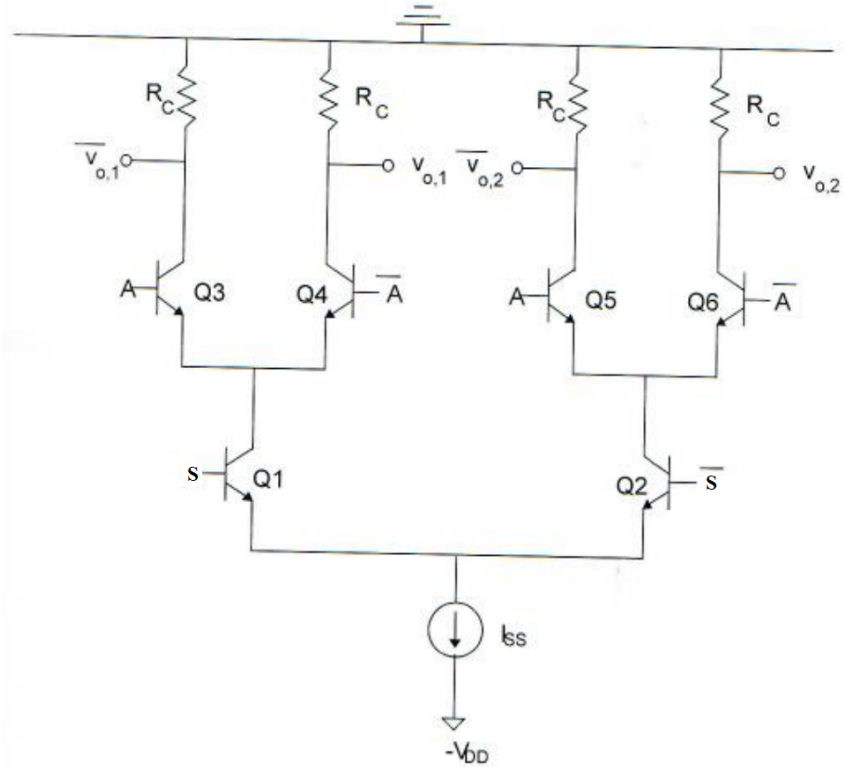
$$V_{IL} \cong V_R - V_T \ln\left(\frac{V_{SWING}}{V_T} - 2\right)$$

- b) Sketch the VTC with a single input change (scenario described above), ranging from - 5.2V to 0, labelling V_{OH} , V_{OL} , V_{IH} , V_{IL} .
 - i. Plot both V_{O1} and V_{O2} by hand.
 - ii. Plot both V_{O1} and V_{O2} with LTSpice. Compare with hand results.
- c) Find the noise margins, NM_H and NM_L for V_{O1} only.
- d) What is the logic function at nodes V_{O1} and V_{O2} ?
- e) Find the fan out N , when $NM'_H = NM_H/2$.
 - i. Find for output V_{O1} only (V_{O2} has too high of a voltage to be a valid input to another gate).
 - ii. Verify fanout at V_{O1} with LTSpice. Compare with hand results.
 - 1) Perform check by verifying that input current matches calculations (add a voltage source to the input, check current the source provides), and that adding a current source to the output degrades the output by the expected amount.
- f) Find the power consumption of this circuit.
 - i. Verify with LTSpice.

Question 2:

Objectives:

- CML Circuit Function



- Find the Boolean expression for the logic function above.
- What type of circuit is this?

Report Submission:

Submit your work on LMS under **Labs**

- Show your work.
 - Include images of your hand calculations with work and final results.
 - Include screenshots of LTspice with all circuit schematics with all output plots, **clearly showing all hand calculated parameters.**
- Due date: Friday at 11:59pm.

Report Submission:

Submit your work on LMS under **Week 12/Lab 10**.

- Show your work.
- Include images of your hand calculations with work and final results.
- Include screenshots of LTspice with all circuit schematics with all output plots, **clearly showing all hand calculated parameters.**
- Due date: Friday at 11:59pm

